AIML SESSION 08 ASSIGNMENT SUBMISSION

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Domain/Subject: AIML (Artificial Intelligence & Machine Learning)

College: Government Polytechnic College Washim

Branch: Information Technology

Session: 09

Q1. Creating Dictionaries)

Write a program to create dictionaries using different methods:

A) An empty dictionary using two different ways.

B) A dictionary with string keys (name, city, course).

C) A dictionary with integer keys.

D) A mixed data type dictionary (string, int, float).

Print each dictionary and its type. Add comments explaining the syntax.

Output Screenshot:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 09/Q1.py"
d1 = {}
Type: <class 'dict'>
d2 = {}
Type: <class 'dict'>

d3 = {'name': 'Vaibhav', 'city': 'Washim', 'course': 'AI-ML'}
Type: <class 'dict'>

d4 = {1: 'India', 2: 'China', 3: 'America', 4: 'Japan'}
Type: <class 'dict'>

d5 = {'Brand': 'TATA', 'Wheels': 6, 'Distance': 129.0}
Type: <class 'dict'>
>
```

Q2. (Accessing and Modifying Elements)

Create the dictionary:

student = {

"name": "YourName",

"age": 00,

"city": "YourCity",

"marks": 00

}

- **Print the value of "name" using square brackets.**
- **Update the "marks" to 92.**
- **Add a new key "grade" with value "A".**
- **Print the updated dictionary.**

Output Screenshot:

Q3. Create a dictionary:
person =
{"name": "Priya", "age": 21,
"profession": "Engineer"}

Demonstrate:

A) Using gets () to access "age" and a non-existing key "salary" (with default value).

B) Print all keys using keys ().

C) Print all values using values ().

D) Print all key-value pairs using items ().

Output Screenshot:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 09/Q3.py"
21
Not Found
dict_keys(['name', 'age', 'profession'])
dict_values(['Priya', 21, 'Engineer'])
dict_items([('name', 'Priya'), ('age', 21), ('profession', 'Engineer')])
>
```

Q4. (Nested Dictionary)

Create a nested dictionary to store information of two students:

Python

students = {

"s1": {"name": "Rahul", "age": 20, "marks": 88},

"s2": {"name": "Sneha", "age": 21, "marks": 95}

}

Print the details of the first student.

Print the marks of the second student.

Add a new subject "math" with marks 90 for the first student.

Output Screenshot:

- **Q5. (update () and pop ())**
- **Create a dictionary: info = {"brand": "Samsung", "model": "A52", "price": 25000}**
- **Update it with new information: {"color": "Black", "price": 24000}.**
- **Remove the key "model" using pop () and print the removed value.**
- **Print the final dictionary.**
-

Output Screenshot:

Q6. (popitem () and clear ())

Create a dictionary with at least 5 key-value pairs.

Use popitem () twice and print the removed items each time.

Clear the entire dictionary using clear ().

Print the final result.

Add comments explaining the difference between pop () and popitem ().

Output Screenshot:

Q7. (copy () and setdefault ())

Create a dictionary: d = {"a": 1, "b": 2}

- **Make a copy of it.**
- **Use setdefault () to add key "c" with value 3 (if it doesn't exist).**
- **Use setdefault () again for an existing key "a".**
- **Print the original and copied dictionaries.**

Output Screenshot:

Q8. (fromkeys () and Membership)

Write a program that:

- **Uses dict. fromkeys () to create a dictionary with keys ["name", "age", "city"] and default value None.**
- **Takes user input to fill these keys.**
- **Checks if a given key exists using in operator.**

Output Screenshot:

Q9. (Practical Application)

Write a program to store and manage contact information using a dictionary:

- **Take name, phone number, and email as input.**
- **Store them in a dictionary with name as key.**
- **Allow the user to search a contact by name using get ().**
- **Print all contacts using items ().**

Output Screenshot:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 09/Q9.py"
Enter Name: Vaibhav
Enter Phone Number: 9898809899
Enter Email: vaibhavkate1301@gmail.com

Enter name to search: Vaibhav

Contact Found:
Phone: 9898809899
Email: vaibhavkate1301@gmail.com

All Contacts:
Name: Vaibhav
Phone: 9898809899
Email: vaibhavkate1301@gmail.com

>
```

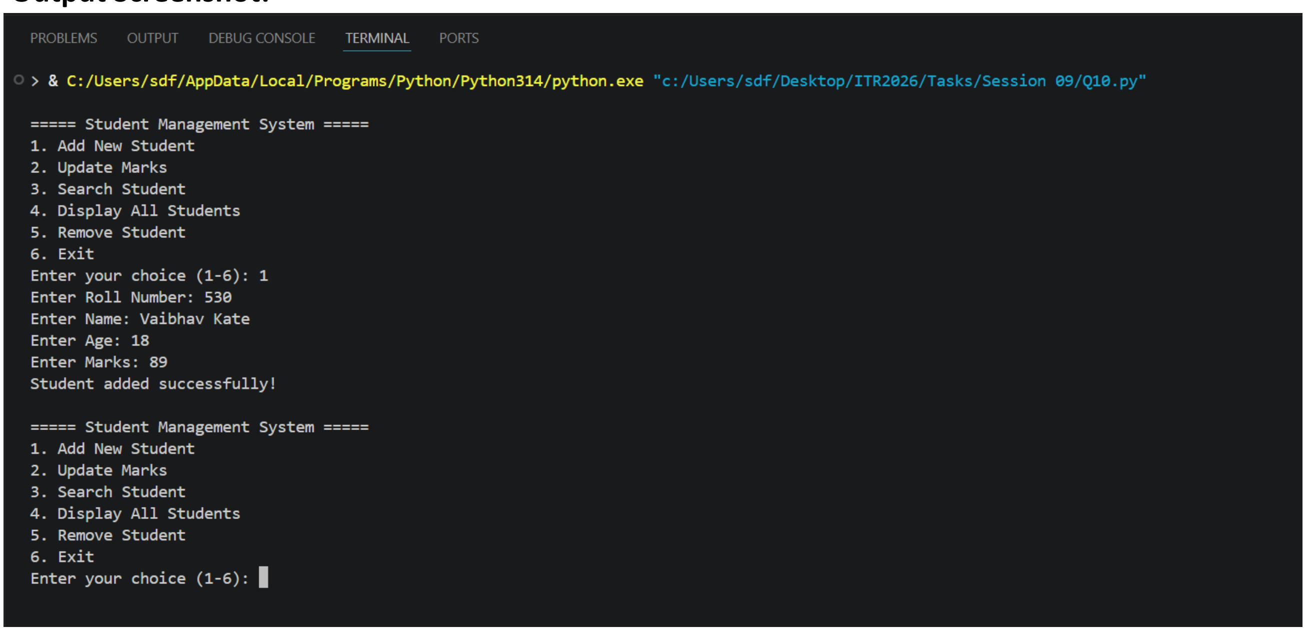
Q10. Mini Project - Combined Concepts)

Create a Student Management System using a dictionary:

- **Repeatedly ask the user to enter student details (roll number as key, and a nested dictionary containing name, age, and marks).**
- **Use a while loop with a menu:**
  1. **Add new student**
  2. **Update marks of a student**
  3. **Search student by roll number**
  4. **Display all students**
  5. **Remove a student**
  6. **Exit**

Use appropriate dictionary methods (get (), update (), pop (), etc.). Add meaningful comments.

Output Screenshot:



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 09/Q10.py"

===== Student Management System =====
1. Add New Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit
Enter your choice (1-6): 1
Enter Roll Number: 530
Enter Name: Vaibhav Kate
Enter Age: 18
Enter Marks: 89
Student added successfully!

===== Student Management System =====
1. Add New Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit
Enter your choice (1-6):
```